CHAPTER 4

Bank Money Creation

This section outlines the process of money creation via the extension of private bank credit. The money so created is known as credit money, or inside money. The vast bulk of money used in modern societies is created in this way, rather than through the actions of central banks. However, as will be outlined below, this money is convertible into central bank money, including bank notes, making it virually economically equivalent. To reflect current practice, electronic book-entry money is used here instead of the historical private bank notes.

The presentation here is based on double-entry book-keeping to make it easier to understand. The double entry principle means that every increase in an asset is accompanied by an increase in a liability or decrease of another asset by an equal amount. In the same way, an increase in a liability required an increase in an asset, or decrease of another liability. As usual, assets are shown on the left and liabilities on the right.

4.1 SINGLE-BANK INTRODUCTION

The essential process in creating money is for a bank extend a loan to a customer by crediting the customer's current account with the loan amount. The loan is a new asset for the bank and this increase of assets is accompanied by an increase in deposit liabilities. The process is shown in Figure 4.1.

It should be noted that, for now, this process has no impact on any economic actor aside from the bank and its customer. The loan exists only as a bilateral contractual relationship between these two entities, as does the balancing deposit.

The nominal amount of the deposit need not be the same as that of the loan. For instance, a Danish mortgage is created by the sale of a new mortgage bond so that the deposit amount is linked to the *market* price of that security at the time. The borrower is responsible for the repayment of his or her share of the bond at its *nominal* value[1].

The key to understanding why money has been created lies in the business model of a bank. Because deposits can readily be converted into cash or transferred, they are effectively means of payment. Bank deposits have a certain degree of credit risk but for most bank customers, this risk is outweighed by the convenience of not having to look

[1]The impact of the loan on the balance sheet of the bank is also simplified here because the bank may recognise certain fees immediately as profit and loss so that they go into the equity portion of the right-hand side of the bank's balance sheet.

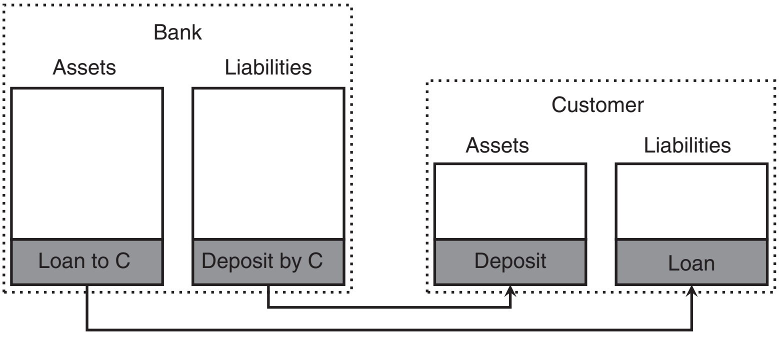

FIGURE 4.1 The money creation process. A bank lends money to a customer and credits this money to the customer's current account. The arrows link corresponding assets and liabilities on the balance sheets of bank and customer.

after and handle physical cash. Still, while bank deposits are denominated in currencies that are legal tender (such as euros, dollars or yen), they are not identical to cash.

In practice, most modern economies restrict this process through reserve requirements. Such requirements oblige the bank to hold central bank money in an amount that covers a certain minimum fraction of a subset of their deposits. Historically, the reserve of a bank was the amount of physical specie it held against redemption requests of bank notes it had issued. Because physical specie is difficult to increase in times of high redemptions (which usually coincide with crises of confidence), reserve holdings had to have a prudent size. In modern economies, central bank money is predominantly electronic and can be created instantly. Reserve requirements are therefore now mainly a means for the creation of a structural demand for central bank money, and therefore a monetary policy tool.

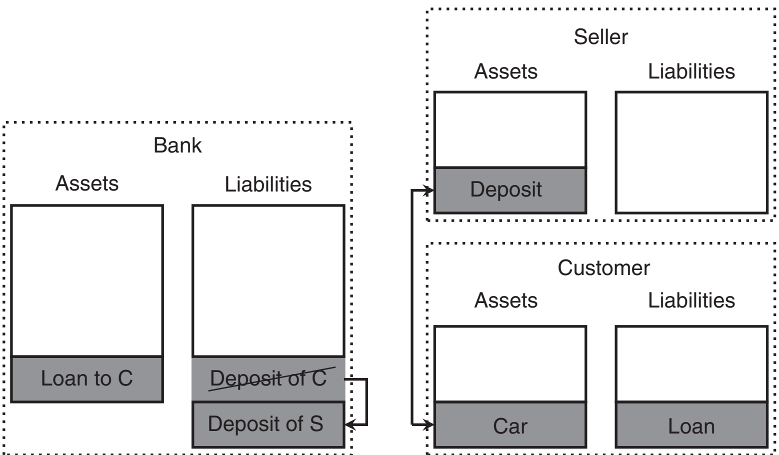

FIGURE 4.2 Transfer of a deposit between two customers of the same bank as part of a sale transaction.

In the simplest case, the customer can use the deposit to purchase an asset, like a car, via a transfer to a seller who has a current account at the same bank. The purchase reduces the bank liability linked to the customer desposit but increases the deposit liability to the seller by the same amount (cf. Figure 4.2).

While this transaction is simple and, as far as money is concerned, only involves the manipulation of book entries in the bank, it has an important precondition: Both customers of the bank need to be willing to transact in the money of this bank, and in this case have current accounts with it. This is a very strong condition and unrealistic in practice. The logical next step is therefore to consider transactions involving multiple banks.

4.2 EXTENSION TO MULTIPLE BANKS

Deposits at different banks, even in the same currency, are at first sight non-equivalent. Since deposits are private current liabilities where the debtor is a bank, they carry the name-specific credit risk of the bank they are held at. Seen as assets, anyone asking somebody else to accept money deposited in a bank for payment in effect asks the payee to accept that bank's credit risk. This includes transfers between deposits at different banks: a customer asking to credit an account with money held at a different bank asks the receiving bank to accept the credit of the bank the holds the original deposit. In the days of private-issue bank notes and cheques, a customer paying a banknote issued by one bank, or a check drawn on it, into an account held at a different bank would in effect ask the receiving bank to accept the credit of the issuing bank.

As discussed in the context of deposits at a single bank, the reason why deposits at different banks in the same currency are largely fungible is that each bank stands ready to pay out deposits in legal tender. A natural limit to the payments a bank can expect to make is that many customers keep money at the bank for a considerable amount of time as part of their savings. A second factor is that banks receive payments as well as make them. This factor leads to the idea of netting.

The amount of interbank credit that customer transfers force banks to extend can be reduced through netting. In simple bilateral netting, two banks A and B work out the net flow between them, i.e., the positive difference of money sent to B from A on one hand, and from A to B on the other. The bilateral flow would then be reduced to the difference of these amounts. More netting efficiency can be obtained through multilateral netting because then flows through intermediaries can be included in the netting calculation. Assuming four banks A, B, C, and D take part and f_{AB} denotes the money sent from A to B, and so on, one can set up a matrix of flows and net across all payment flows as shown in Figure 4.3.

This type of netting was indeed performed for paper cheques by placing cheques on a large table subdivided in the same way as the matrix in Figure 4.3. Because the reduction of payment obligations is a joint benefit of the participating banks, they have an interest to set up and optimise this form of cooperation despite competing for the underlying business that creates the payment obligations in the first place. Netting is more effective when done over a larger volume of payments. This argues for holding

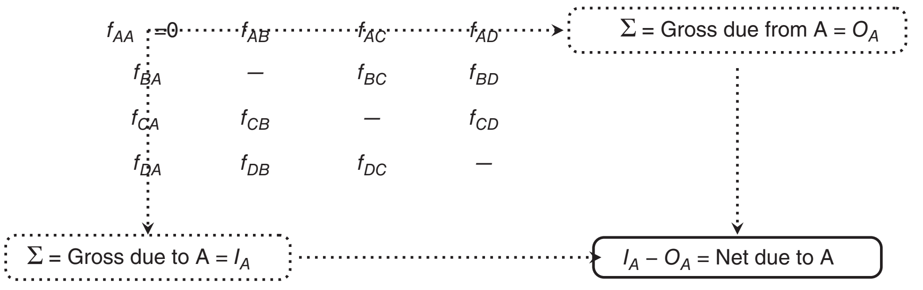

FIGURE 4.3 Netting across multiple interbank transfers. The final receivable or payment obligation of bank A is calculated by first summing up inflows and outflows separately and then netting the result. The net payments of all banks add to zero.

payments for some time, netting them and then processing the net amounts. A longer netting period, however, allows for more imbalances to build up which increases risk in the system. However, even after netting is performed and interbank payment obligations are reduced in this way, the net amounts will not be zero at the end of each day. Over time, payments between banks should more or less balance out, but that does not reduce the need to process payments.

It is therefore still necessary to have mechanisms that allow a change of obligor through interbank transfers. There are two ways to arrange this, both of which are in common use. One is based on bilateral relationships where banks hold deposits for each other, the other involves central clearing banks. The latter can be central banks, and this is indeed the form of interbank transfer that is the most prevalent in developed economies. As will become clear, the second type of transfer is simply a special form of the first.

The first transfer mechanism uses so-called nostro accounts which are current accounts held by one bank at another bank[2]. When bank A is asked to make a payment to a customer S banking at bank B where bank A holds a sufficient amount of money in a nostro account, it will ask B to transfer money from that nostro account into the account of S. Viewing this in terms of the four balance sheets affected, the following picture emerges:

Customer C The deposit (asset) at bank A is reduced by the transfer amount, this is offset by the inflow of whatever asset has been purchased from S.

Customer S The deposit (asset) at bank B is increased by the transfer amount, this is offset by the outflow of whatever asset has been sold to C.

Bank A The deposit of A (liability) is reduced by the transfer amount, this is offset by the reduction in the nostro account balance (asset) by the transfer amount. The size of bank A's balance sheet contracts by the transfer amount which is intuitive because the transfer is an outflow of assets.

Bank B The nostro account balance of A (liability) is reduced by the transfer amount, this is offset by an equal increase of the deposit of S (liability).

[2]'Nostro', meaning 'ours', refers to the account holding bank viewing it as their asset held somewhere else.

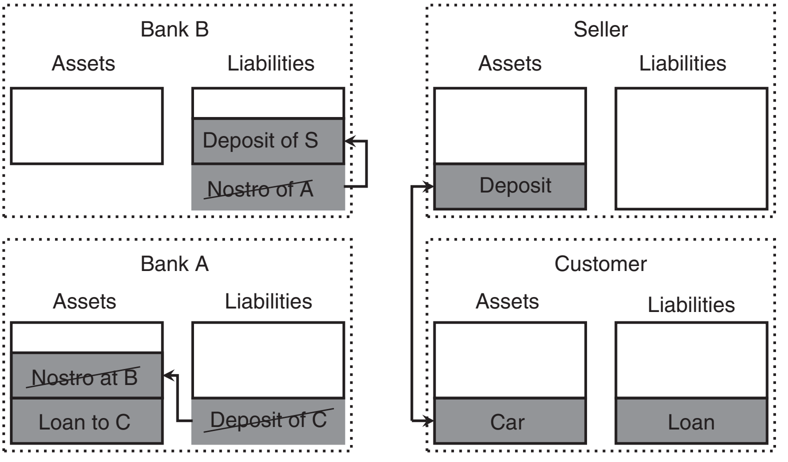

FIGURE 4.4 Transfer of a deposit between two customers of two different banks using a nostro account. For bank B, this transfer is an internal transaction between the deposits of two customers (S and bank A).

The transfer is depicted graphically in Figure 4.4.

An important question is how nostro account balances are created in the first place. These balances are inside money, which is in this case created by bank B. A simple way to create such money is for bank A to simply ask for a loan from bank B. Such a loan would usually be collateralised in some form so that when bank B discounts (cf. Section 4.4.2) a bill drawn on bank A, B does not accept unprotected the credit risk of bank A. Alternatively, bank A could sell an asset, or a debt obligation to the public and deposit the funds so raised in its nostro account at bank B. last but not least, bank A can receive payments into its nostro account at B if the reverse transaction to the one presented here is executed between customers like C and S.

Nostro accounts used in this way pay an important role in cross-border finance because not every bank will have an equal presence in every market. If bank A is active in one part of the world but would like to offer regular payment services to its clients in a different jurisdiction, it can do so by engaging the services of a bank B active in that jurisdiction as explained here. For instance, even a small manufacturer would typically need to pay suppliers and receive payments from costumers in other countries. As it would be impractical for the manufacturer to maintain accounts at banks in every jurisdiction involved, or to bank with a bank that is active in all of them, it will instead use a bank that has correspondent banking relationships to process these payments[3].

Nostro accounts have the disadvantage of consuming balance sheet for the banks involved. They need to hold sufficient funds to deal with usual cumulative flows

[3]It should be mentioned that correspondent banking is not without problems. Because the correspondent bank makes payments on behalf of another bank without having full knowledge of the underlying transaction and (in the example here) one of the customers involved (C), it may unwittingly aid the flow of illegally obtained funds.

between two banks and these funds lie idle unless, and until, there is a customer payment requiring their use. Banks can reduce idle balances in nostro accounts by regularly sweeping them into other assets but the balance sheet benefit of doing so is still limited.

Instead of settling interbank payments bilaterally between the two banks involved, a more efficient mechanism has arisen where third-party payment processors centralise the payment flows between sets of banks. In the UK, these central payment processors are large private banks known as money-centre banks. The German savings banks use regional central banks (Girozentalbanken and Landeszentralbanken) that are jointly owned by the participating savings banks and their regional governments. The smaller Japanese banks use their jointly owned respective central treasuries (Norinchukin, Shinkin Chukin, Shoko Chukin, etc.)[4].

The central payment processor can be seen as a bank in which every participating bank has a nostro account. The net payments arising after a netting process as shown in Figure 4.3 are executed by crediting and debiting these nostro accounts. In this arrangement, instead of every bank having to maintain a nostro account at every other participating bank, it only maintains one account at the central payments processor.

It follows from this setup that the central payments processor must have a superior credit relative to that of the participating bank because its central role requires that every participating bank treats a deposit at the central processor as free of default risk. The standard way of ensuring this superior credit is to restrict the activities of this institution or equip it with strong guarantees of a highly trusted entity. Today's financial systems rely on central banks to perform this role so that the mechanism can be explained in that context.

4.3 TRANSFER SETTLEMENT IN CENTRAL BANK MONEY

The final step in the evolution of modern payments processing is the merger of the central payments processing function with that of a central bank (central banks are discussed in detail in Chapter 5). A deposit at a central bank is money and there is therefore no credit risk associated with such a deposit[5]. The larger central banks offer interbank payments systems that enable banks to transfer deposits held at the central bank between each other. Banks may choose to conduct netting as discussed in

[4]In the German system in its prime, the savings bank in one region used their regional Landesbank to process payments between them, while payments between regions were handled by a central bank for the Landesbanken. The successor of this bank is now known as DekaBank Deutsche Girozentrale. Its central role of payments processing has become less important, however, through the advent of the ECB's TARGET2 system. DekaBank now handles mostly central securities-related tasks and the German payments system has evolved from a three-tiered to a two-tiered structure.

[5]It should be remembered, however, that credit risk in this sense refers to the risk of money not being available when called for. Even a central bank deposit carries the risk that the money it refers to loses acceptance in the private market, i.e., the risk of a currency devaluation or capital controls.

TABLE 4.1 The main real time gross payments systems.

Central bank	Name	Access
Federal Reserve	FedWire	Federal Reserve member banks, depositary intitutions, foreign bank branches
ECB	TARGET2	Eurosystem counterparties (MFIs)
Bank of Japan	BOJ-NET	Banks (banks, trust banks, shinkin banks, etc.), branches of foreign banks, securities firms, securities finance firms, money market brokers
Bank of England	RTGS/CHAPS	Banks, building societies, investment firms and central counterparties
People's Bank of China	High-value payment system	Banks, PBoC branches

Figure 4.3 but such netting, if it occurs, is of no concern to the central bank. Transfers between current accounts at a central bank are executed as instructed, which implies that they are viewed as gross transfers by the central bank. To enable efficient use of the deposits, transfers are executed in real time or near real time.

The essential difference between using central payments processors and central banks for interbank payments is that accounts held at a central bank are economically identical to legal tender. Payment in central bank money, or transfers between central bank current accounts, therefore achieve immediate finality, i.e., transfers of assets that are not subject to the credit risks of the parties involved.

On the other hand, when the central payment processor is a bank then the liquidity generated in this bank (in other words, the credit balances held by the participating banks) is potentially cheaper than central bank liquidity. Jointly owned clearing banks would distribute interest earned on debit balances (after costs) to their owners, which are at the same time the users of the system. If the same users had to source central bank cash at the prevailing policy rate, the same interest earned at the central bank would be paid out to someone else (usually the government). As mentioned above, the downside is that the participants assume convertibility risk, i.e., the risk that the clearing bank fails and balances held there cease to be convertible into central bank money. Usually, therefore, the joint owners of such clearing banks also guarantee them jointly.

Some payments systems provide intraday credit, which in essence means that while in principle only transfers between positive account balances can be made through these systems, the central bank providing such accounts allows them to be overdrawn during the day.

The practical reason to allow such overdrafts is to make the system robust against the reordering of transactions. Table 4.2 shows an example of how the ordering of transactions can determine whether an overdraft is required or not.

Incentivising payment participants to re-order transactions in order to avoid the recourse to intraday overdrafts can be counterproductive. Every participant would seek to delay outgoing payments and accelerate incoming flows although the incoming flow for one participant is an outgoing flow for another.

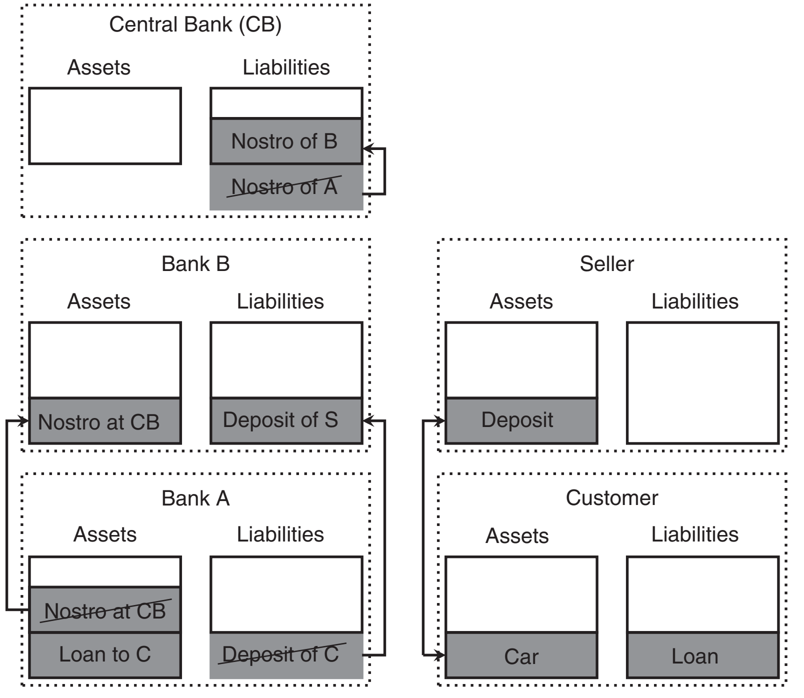

FIGURE 4.5 Transfer of a deposit between two customers of two different banks using a central clearer or central bank. In essence, part of the balance sheet of bank A is transferred with finality to the balance sheet of bank B. The central bank balance sheet size is unaffected because only the designation of an existing liability changes.

TABLE 4.2 Example for the effect of transaction ordering on intra-day cash balances. Changing the timing of the transaction marked in bold makes the difference between requiring an intraday overdraft or not.

Case A			Case B		
Time	**Payment**	**Balance**	**Time**	**Payment**	**Balance**
6:00		100	6:00		100
8:15	50	150	8:15	50	150
9:23	**–200**	**–50**	10:01	100	250
10:01	100	50	**11:04**	**–200**	50
12:03	50	100	12:03	**50**	100
19:00		100	19:00		100

At the same time, allowing overdrafts exposes the payment system to the risk that a participant turns out to be unable close a negative balance at the end of a payment day. This is the reason why direct participation in central bank operated payments system is generally restricted to monetary policy counterparties. Such institutions have in any case recourse to funds-providing facilities of the central bank that grants the overdraft.

Any negative balances left at the end of a trading day can be covered with new central bank money. However, the recourse to such a facility may be seen as negative by the central bank and be sanctioned. System participants would normally be expected to arrange for end-of-day balances to be covered by normal means, including regular funding operations by the central bank.

As a result, payment systems without overdraft can encourage higher cash holdings at participating banks. In essence, these banks substitute overdrafts at the payment system with higher cash reserve holdings. From a systematic point of view, the intraday liquidity provision is done in the private sector, rather than at the central bank. Whether this is preferrable is in essence a matter of belief. From a monetary policy viewpoint, the transactional money demand is a distraction from the central banks stance. Having to accommodate such demand visibly through overnight cash, rather than implicitly through intraday credit in the payment system, can therefore sometimes conflict with the intended signalling effect. In the US, where the Fed normally targets the overnight rate through daily open market operations, this downside is less pronounced than in the euro area, where the main pre-crisis operation was a weekly repo tender. This may be one reason why TARGET2 with intraday overdrafts and FedWire without this facility form stable equilibria with the respective operational frameworks.

4.4 TRADE AND NON-BANK CREDIT

Trading as a source of lending in the economy is often neglected in macroeconomic analyses. Most of the trade between companies, and sometimes between companies and customers, is conducted on credit. Sellers of goods and services tend to invoice customers in arrears. This is related to the idea that a trader buying a good needs some time to resell it before the maker could be paid from the proceeds. From an economic viewpoint, delivering a good or service for later payment amounts to a loan granted by the producer to the trader. The terms of this loan, in particular the currency and duration, can be negotiated freely between the trading partners.

The extent of trade credit can be measured on a single company level by the accounts payable and accounts receivable items on corporate balance sheets, at least on reporting days. By comparing the sales of a company (i.e., the flow of receivables through the company's accounts) to its accounts payable balance sheet item, one can calculate the the average time between finalising a sale and eventual payment. This is known in corporate finance as the collection period [13, 46]:

$$\text{Collection period} = \frac{\text{Accounts receivable}}{\text{Total sales}} \tag{4.1}$$

Note that accounts are usually published on a quarterly basis and corporations may have incentives to manage balance sheet items up or down around reporting dates. For instance, a company might prefer to be paid before December 31st on a particular invoice in order to reduce accounts receivable at year end. Doing so would shorten its collection period as defined in Equation (4.1). In other circumstances, leaving an outstanding balance with a customer may offer a better return than depositing cash with a bank and a later payment would be preferable.

Although most markets have established conventions on invoice currencies and payment terms, there is a considerable degree of freedom for corporate treasuries to manage the currency and maturity composition of their accounts payable and receivable. Deviations from market standards may incur extra costs but may be outweighed by the benefits, whether it be through appreciation by users of published accounts or through economic returns. As a result, terms of payment are not simply a matter of convention but can be part of active balance sheet management.

It is even possible to vary invoice periods by currency. A company that pays invoices quickly in currency A but allows for invoices in currency B to be paid to itself over longer periods replicates a short A, long B position. Such a position can be difficult to spot in corporate accounts, and therefore in international balances of payments.

4.4.1 Non-cash trading instruments

Trading on credit can take the form of trading on account or the form of negotiable non-cash instruments such as bills of exchange and cheques. When trading on account, the seller maintains an account for each customer. Deliveries to the customer lead to debits against the account while payments from the customer are credited to the same account. Accounts can be in surplus (as a result of overpayment by the customer) but are usually in debit or flat. Trading on account is common when traders interact with non-traders because the relationship is strongly asymmetric.

Bills of exchange are payment instruments that represent an unconditional obligation of the party that it is drawn on, the drawer. Bills are payable on demand after their stated first payment date on presentation to the drawer. Usually, the seller of a good would issue a bill and present it to the buyer for acceptance in exchange for handing over the goods. Once accepted, the bill becomes unconditional. Due to this property bills themselves are being used as means of payment between traders. The payee of a bill can transfer it as payment to another trader by means of endorsing it to the recipient. For bills in paper form, this involves a simple signature on the back of the bill. Endorsement does not change the payment amount or payment date of the bill. However, it creates a conditional liability on the trader that endorsed it. If the original drawer of the bill refuses to, or is unable to, pay the bill as specified, the holder of the bill can seek redress from the trader who endorsed the bill. As such, by using a bill drawn on a third party as payment does not release a trader from a final obligation to pay in full for goods or services received.

If a bill is accepted by a banker on behalf of the ultimate payee, it is called a Banker's acceptance, or BA. By accepting the bill, the banker acts as both agent to the acceptor and, usually, guarantor to the issuer of the bill. A cheque is a payment instrument drawn on a bank subject to sufficient funds in the account of the payer. Cheques can be post-dated to prevent conversion into cash before a set date. As such, a banker's acceptance and a cheque are similar in that a bank provides additional security to payees accepting them for payment. They differ in who is the legal obligor and in the conditionality of payment.

4.4.2 Discounting

A trader can sell a bill to a banker for an immediate cash payment. This is known as discounting and represents one of the oldest forms of bank credit as well as the origin of the word discount in finance. When the banker buys the bill, the cash amount paid will normally be lower than the face value of the bill. The ratio between the two is known as the discount factor:

$$\text{Discount factor} = \frac{\text{Cash payment}}{\text{Face value}} \tag{4.2}$$

The discount factor is usually expressed in the form of an interest rate, known as a discount rate.

A bank that has discounted a bill may re-sell it to another bank. This is known as rediscounting. Typically, central banks used to inject liquidity through rediscounting (see Section 7.2).

4.4.3 Delineating payment instruments from money

Traders can use bills and cheques drawn on third parties to pay each other which raises the question whether there is a difference between money and these non-cash payment instruments.

As usual in economics, the answer is complex. In corporate finance, bills held are separated from the balance sheet category of 'Cash and Cash Equivalents' on the asset side but both of them are counted towards the wider category of 'Current Assets'. Monetary aggregates do not count bills of exchange or cheques as money but this may simply be due to the difficulty in measuring the relevant outstanding amounts. Note that no bank is able to quantify the nominal amount of cheques drawn on it until they are presented for payment.

However, when a bill or cheque becomes payable, they do represent claims payable in money. This property is what enables them to be used in transactions. A number of differences between non-cash payment instruments are noteworthy although they do not explain their omission from monetary aggregates:

No legal tender Payees can refuse to accept bills or cheques as payment for debts payable. This is one of the motivations for discounting a bill, namely the operation of transforming a bill into legal tender, subject to an appropriate discount.

Immediacy A bill is payable by the drawer upon presentation. Unlike cash, this delays payment. The payee has to present the bill at a given location at a given time to receive cash. This cash may in any case be refused and would have to be recovered through redress from intermediate endorsers of the bill.

Limitations Bills normally cannot be claimed beyond a certain period after their due date (usually one year). Cash, in contrast, has no such limitation.

4.5 DIGITAL TOKEN MONIES AND CRYPTOCURRENCIES

In recent decades, virtual currencies known as cryptocurrencies such as Bitcoin or Ethereum have been created. Such currencies not only do not have a nominal anchor, but are not even tied to any given jurisdiction. The are used as means of exchange through distributed databases that allow anonymous and nearly instantaneous[6] transfers between participants. The way in which cyber-currencies seek to achieve the store of value property required of money is by tying the creation of currency units to complex mathematical calculations which have a meaningful execution cost (so-called proof-of-work problems). Overall, cryptocurrencies are more than new currencies. They depend for their functioning on a complex ecosystem of users willing to provide storage and computing power, mathematical laws that guarantee a limited supply, and protocols that solve attendant problems, such as how to prevent the same crypto-asset being spent twice in two separate transactions [62]. The reason why Bitcoin succeeded as well as it did is that the economic incentives for all participants were carefully balanced, or rather, the Bitcoin incentives were balanced in a way that led to its success.

Cyber-currencies replicate the salient features of cash transactions through cryptographic means. In general, this is achieved through the use of a cryptographically safe public transaction log that is replicated across the entire network of participants in which every single transaction is recorded (the so-called blockchain). Transactions are considered final once they are recorded in the blockchain for a sufficient amount of time; participants are anonymous because they are only identifiable by the public part of their asymmetric keys, and the replication of the blockchain removes the need for central recording of monetary assets, such as bank accounts.

The main criticism of cryptocurrencies is that they expend a very large amount of physical assets (electricity and specialised hardware) on achieving aims that in the majority of cases could be achieved in much less elaborate ways. Immediate finality of payment combined with a reasonable degree of anonymity and decentralisation has been achieved in commercial settings through pre-paid electronic monies such as Suica in Japan (2001) long before cryptocurrencies became popular. Some technical concepts used in cryptocurrencies, such as the blockchain, are interesting in other settings and may speed up certain transaction types in other contexts. However, a blockchain does not create a currency and digital payments can be handled, and have been handled for many years, without the use of a blockchain.

Digital token currencies are substantially different from cryptocurrencies because they do not seek to replace the entire institutional infrastructure of modern monies and may even be issued by central banks. As a concept, these monies replace classical physical tokens, such as numbered bank notes, with digital equivalents. The transfer of such tokens can be organised through centralised or distributed databases depending on which system provides better usability and cost efficiency.

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[6]In practice, money transfers in cryptocurrencies appear to be instantaneous but are in fact not final until some considerable time (up to several hours) later. This compares to near immediacy of bank-based transfers in most developed and some developing nations.

4.6 THE MONEY MULTIPLIER

Because bank notes are debt, there is in theory no limit as to how much of them (in terms of nominal amounts) can be issued by a given bank. This is unlike certificates where the total amount in circulation is, by definition, equal to the physical amount of physical commodity money in storage[7]. Because bank notes are sometimes cashed, issuers needed to hold a certain amount of physical metal against such repayment demands. This amount of physical cash is called the reserve. For reasons discussed below, bankers have an economic interest to hold as small a reserve as possible, but prudence suggests as large a reserve as possible. No matter how large an actual reserve is, there is always a risk of more bank notes being presented for payment than the size of the reserve. Such a large repayment demand is known as a bank run[8].

The economic importance of banking is less the creation of bank notes than the lending of the deposits. By doing so, the bank in effect creates new money because not only are the bank notes in circulation, but also the cash deposit that the bank has lent to a third party. Furthermore, the cash that has been lent on can later end up as another deposit, i.e., new bank notes. To give a simple example, imagine 10 gold coins being deposited with a bank. The bank issues a 10 coin note against it. Assuming that the bank generally holds a 10% reserve against deposits, up to nine coins can be lent by the banker to a borrower. The borrower may purchase some goods with this money and the seller of those goods could deposit the nine coins against a bank note for nine coins. The total amount of money now in circulation is equivalent to 19 coins, namely the two bank notes of 10 and 9 coins. Again the banker can lend part of the deposit (8.1 coins given the 10% reserve), which may again be deposited and so on, ad infinitum. In total, the maximum amount of paper money that can be based on the 10 physical coins is equivalent to 100 coins. Mathematically, the maximum amount of money M that can be based on an initial deposit of one coin with a reserve ratio of r is given by:

$$M = \lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{(1+r)^k} = \frac{1}{r} \tag{4.3}$$

The number M is known as the monetary multiplier and the coins in the example are known as high-powered money because an increase in the number of coins leads to an M-fold increase in the amount of money in circulation. The limit $n \to \infty$ suggests that the number M is theoretical, but although the actual amount of money in circulation may be lower than suggested by the theoretical M, the relationship between this amount, the amount of high-powered money, and the reserve ratio r, is intact. Not least, this is because there is no reason for loans to be paid in coins. The bank could

[7]Perhaps the most well-known remaining certificate money is the paper money used in Scotland, Northern Ireland, and the Channel Islands. Aside from small 'authorised issues', these bank notes issued by private banks are backed by identical stores of English pound notes specially issued by the Bank of England. These special issues have a denomination of one million pounds.

[8]One way to reduce the risk of bank runs is to create a lender of last resort, cf. Chapter 5.

easily create 100 coins worth of notes upon the deposit of 10 coins, and use 90 of these for lending.

However, interbank payments normally have to be settled in specie (or, in modern times, central bank money) instead of the bank notes issued by the banks themselves. This requirement for settlement money creates an additional limit on the money multiplier M. In practical terms, therefore, the money multiplier has little relevance.

The problem of bank runs has led to the establishment of banks that act as a lender of last resort to banks subject to a run. The banks in London deposited part of their reserves with the Bank of England (against interest) with the understanding that the Bank would lend the accumulated reserves to banks that were subject to a bank run [5]. Although the total bank reserves were much lower than the total amount of bank notes in circulation, any one bank could survive a bank run by borrowing from that central repository. A similar system was established as a reaction to bank failures in the 1920s in the United States in the form of the Federal Reserve Bank System.

The pooling of physical specie (which after all, incurs a cost and pays no interest) reduces the cost of maintaining a sufficient reserve compared to each bank holding a sizeable amount of specie. This pooling is one avenue along which the notion of central banks, discussed further below, has evolved.